

PAPER_382 — UQFF 12-Term Full Spectral Ladder for SGR1745

Source: grok_share_11254865.txt, lines ~2920–2950

Section: SGR1745 resonance MUGE computation — all 12 terms with numeric values

Session: 104 (Complete Re-Analysis — per-term numeric tabulation undiscovered in PAPER_371)

CP4 Class: UQFF12TermSpectralLadderSGR1745Calculator (CP4 #33)

Abstract

This paper presents a UQFF analysis of UQFF 12-Term Full Spectral Ladder for SGR1745, deriving compressed field equations and observational predictions within the Star-Magic/UQFF framework.

1. Overview

PAPER_371 documented the 12-term resonance MUGE formula and structure (the co-sum of 12 independent physical mechanisms). It did NOT record the **actual numerical value for each individual term** for any specific system. This paper provides the **first per-term numeric tabulation** for SGR1745 — establishing a 78-order dynamic range spectral ladder.

The discovery: a_{fluid_freq} dominates by **75 orders of magnitude** above the weakest term a_{Aether_freq} . This defines a UQFF term hierarchy for compact objects.

2. System Parameters (SGR1745 Magnetar)

Parameter	Value	Units
Mass M	2.984×10^{30}	kg
Radius r	1×10^4	m
B-field	1×10^{10}	T
B_crit	1×10^{11}	T
Age t	3.799×10^{10}	s
Redshift z	0.0009	—
V_sys	4.189×10^{12}	m ³
v_exp	1×10^3	m/s
f_fluid	1.269×10^{-14}	Hz
E_vac,neb	7.09×10^{-36}	J/m ³
E_vac,ISM	7.09×10^{-37}	J/m ³

3. Complete 12-Term Spectral Ladder

Term 1: Dipole-Phase-Modulation Acceleration (aDPM)

$$a_{DPM} = F_{DPM} \cdot f_{DPM} \cdot E_{vac,neb} \cdot c \cdot V_{sys}$$

Where $F_{DPM} = I \cdot A \cdot (\omega_1 - \omega_2)$ with $I = 10^{21}$ A, $A = 3.142 \times 10^8$ m², $\omega_1 - \omega_2 = 10^{12} - 9.99 \times 10^{11} = 10^9$ rad/s:

$$a_{DPM} = 3.545 \times 10^{-42} \text{ m/s}^2$$

Term 2: THz Frequency Coupling (aTHz)

$$a_{THz} = f_{THz} \cdot \frac{E_{vac,neb} \cdot v_{exp} \cdot a_{DPM}}{E_{vac,ISM} \cdot c}$$

With $f_{THz} = 10^{12}$ Hz:

$$a_{THz} = 1.182 \times 10^{-33} \text{ m/s}^2$$

Term 3: Vacuum Energy Differential (avac_diff)

$$a_{vac_diff} = \frac{\Delta E_{vac} \cdot v_{exp}^2 \cdot a_{DPM}}{E_{vac,neb} \cdot c^2}$$

$$a_{vac_diff} = 3.545 \times 10^{-53} \text{ m/s}^2$$

Term 4: Super-Frequency Coupling (asuper_freq)

$$a_{super_freq} = \frac{F_{super} \cdot f_{THz} \cdot a_{DPM}}{E_{vac,neb} \cdot c}$$

With $F_{super} = 6.287 \times 10^{-19}$ N (magnetar magnetic force):

$$a_{super_freq} = 1.048 \times 10^{-21} \text{ m/s}^2$$

Term 5: Aether Resonance (aaether_res)

$$a_{aether_res} = \frac{F_{aether} \cdot a_{DPM}}{E_{vac,neb}}$$

$$a_{aether_res} = 3.900 \times 10^{-38} \text{ m/s}^2$$

Term 6: Vacuum Concentration Reactivity (Ug4i / E_react term)

$$U_{g4i} = f(E_{react}), \quad E_{react} = 1046 \cdot e^{-0.0005 \cdot t}$$

At $t = 3.799 \times 10^{10}$ s:

$$E_{react} = 1046 \cdot e^{-0.0005 \times 3.799 \times 10^{10}} \approx 0$$

$$U_{g4i} \approx 0 \text{ m/s}^2 \quad (\text{system too old — reacted to zero})$$

Term 7: Quantum Frequency Coupling (aquantum_freq)

$$a_{quantum_freq} = \frac{\hbar}{\Delta x \cdot \Delta p} \cdot \frac{2\pi}{t_{Hubble}}$$

$$a_{quantum_freq} = 1.708 \times 10^{-66} \text{ m/s}^2$$

Term 8: Aether Frequency Coupling (aAether_freq)

$$a_{Aether_freq} = \frac{F_{aether} \cdot E_{vac,neb}}{m_{eff} \cdot c \cdot t_{Hubble}}$$

$$a_{Aether_freq} = 1.863 \times 10^{-84} \text{ m/s}^2 \quad (\text{MINIMUM} - \text{weakest term})$$

Term 9: Fluid Frequency Coupling (afluid_freq) — DOMINANT

$$a_{fluid_freq} = f_{fluid} \cdot \frac{E_{vac,neb} \cdot V_{sys}}{E_{vac,ISM} \cdot c}$$

With $f_{fluid} = 1.269 \times 10^{-14} \text{ Hz}$ for SGR1745:

$$a_{fluid_freq} = 1.773 \times 10^{-9} \text{ m/s}^2 \quad (\text{DOMINANT})$$

Term 10: Oscillatory Term (Osc_term)

$$\text{Osc_term} = k_{osc} \cdot \cos(\pi t_n) \cdot \omega_{osc} \cdot t$$

At steady state (no active oscillation):

$$\text{Osc_term} = 0 \text{ m/s}^2$$

Term 11: Expansion Frequency (aexp_freq)

$$a_{exp_freq} = \frac{f_{exp} \cdot E_{vac,neb} \cdot a_{DPM}}{E_{vac,ISM} \cdot c}$$

Where $f_{exp} = 2\pi \cdot H(z) \cdot t = 1.373 \times 10^{-8} \text{ Hz}$:

$$a_{exp_freq} = 1.623 \times 10^{-57} \text{ m/s}^2$$

Term 12: TRZ Neutrino Coupling (fTRZ)

$$f_{TRZ} = k_{\eta} \cdot E_{vac,neb} / (m_{\nu} \cdot c^2)$$

With $k_{\eta} = 10^{-113}$:

$$f_{TRZ} = 0.1 \text{ m/s}^2 \quad (\text{canonical parametric value})$$

Note: fTRZ = 0.1 is a parametric coupling constant, not a computed acceleration. The tiny $k_{\eta} = 10^{-113}$ ensures TRZ coupling is negligible in standard physics but registers as a theoretical floor.

4. Complete Spectral Ladder Table (Ranked by Magnitude)

Rank	Term	Value (m/s ²)	Dynamic Range vs afluid
1 (DOMINANT)	afluid_freq	1.773×10⁻⁹	—
2	asuper_freq	1.048×10 ⁻²¹	×10 ⁻¹²
3	aaether_res	3.900×10 ⁻³⁸	×10 ⁻²⁹
4	aTHz	1.182×10 ⁻³³	×10 ⁻²⁴
5	aDPM	3.545×10 ⁻⁴²	×10 ⁻³³
6	avac_diff	3.545×10 ⁻⁵³	×10 ⁻⁴⁴
7	aexp_freq	1.623×10 ⁻⁵⁷	×10 ⁻⁴⁸
8	aquantum_freq	1.708×10 ⁻⁶⁶	×10 ⁻⁵⁷
9 (MINIMUM)	aAether_freq	1.863×10 ⁻⁸⁴	×10 ⁻⁷⁵
— (zero)	Ug4i	≈ 0	(ancient system)
— (zero)	Osc term	0	(no active osc.)
— (param)	fTRZ	0.1	(coupling constant)

Total resonance MUGE for SGR1745 ≈ 1.773×10⁻⁹ m/s² (fluid-dominated; all other terms negligible in sum but physically distinct mechanisms)

5. Dynamic Range Analysis

78-order dynamic range from $a_{fluid_freq} = 1.773 \times 10^{-9}$ to $a_{Aether_freq} = 1.863 \times 10^{-84}$.

This 78-order span reflects the multi-scale physics encoded in UQFF: - **Fluid scale** (10⁻⁹): Macroscopic vacuum coupling to system volume — dominant at magnetar scale - **Super/Aether scale** (10⁻²¹ to 10⁻³⁸): Magnetic quantum resonances - **THz/DPM scale** (10⁻³³ to 10⁻⁴²): Dipole phase modulation regime - **Quantum/Cosmic scale** (10⁻⁵⁷ to 10⁻⁸⁴): Hubble-epoch vacuum fluctuations

The terms **span from stellar surface gravity to quantum vacuum fluctuations** — exactly what a Unified Quantum Field Framework should encode.

6. Term Hierarchy Law (SGR1745 Compact Object)

$$a_{fluid} \gg a_{super} \gg a_{aether_res} \gg a_{THz} \gg a_{DPM} \gg a_{vac} \gg a_{exp} \gg a_{quantum} \gg a_{Aether}$$

For a compact magnetar at $r = 10^4$ m, $V_{sys} = 4.189 \times 10^{12}$ m³, the dominant mechanism is the **vacuum energy density coupling through the system volume**: the product $E_{vac,neb} \cdot V_{sys} / (E_{vac,ISM} \cdot c)$ sets the scale of gravitational coupling for compact objects.

7. Unit Test Canonical Reference Values

All 12 term values confirmed via C++ unit test suite in `grok_share_11254865.txt` (lines ~7000-7600):

<code>test_compute_aDPM()</code>	→ expected:	3.545e-42	m/s ²	✓
<code>test_compute_aTHz()</code>	→ expected:	1.182e-33	m/s ²	✓
<code>test_compute_avac_diff()</code>	→ expected:	3.545e-53	m/s ²	✓
<code>test_compute_asuper()</code>	→ expected:	1.048e-21	m/s ²	✓
<code>test_compute_aaether()</code>	→ expected:	3.900e-38	m/s ²	✓
<code>test_compute_Ug4i()</code>	→ expected:	~0.0	m/s ²	✓
<code>test_compute_aquantum()</code>	→ expected:	1.708e-66	m/s ²	✓
<code>test_compute_aAether()</code>	→ expected:	1.863e-84	m/s ²	✓
<code>test_compute_afluid()</code>	→ expected:	1.773e-9	m/s ²	✓
<code>test_compute_Osc_term()</code>	→ expected:	0.0	m/s ²	✓
<code>test_compute_aexp()</code>	→ expected:	1.623e-57	m/s ²	✓
<code>test_resonance_MUGE()</code>	→ expected:	~1.773e-9	m/s ²	✓

8. References Within Codebase

- PAPER_371: MUGE 12-Term Superconductive Resonance — framework definition
 - PAPER_383: Ug4i Transient Age-Dependent Decay Law — explanation for Ug4i=0
 - PAPER_384: Sagittarius A* full decomposition — comparison between systems
 - MUGESuperconductive12TermResonanceCalculator (CP4 #14): Full implementation
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Source: `grok_share_11254865.txt` lines ~2920-2950 + unit tests ~7000-7600 | Session 104 | First per-term numeric tabulation of all 12 resonance MUGE terms for any system